

PROJECT REPORT

on

**SINE WAVE INVERTER WITH INSTANTANEOUS
DIGITAL CONTROL**

Submitted by

JAUHAR K (20319048)

in partial fulfilment of requirement for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION



DIVISION OF ELECTRONICS ENGINEERING

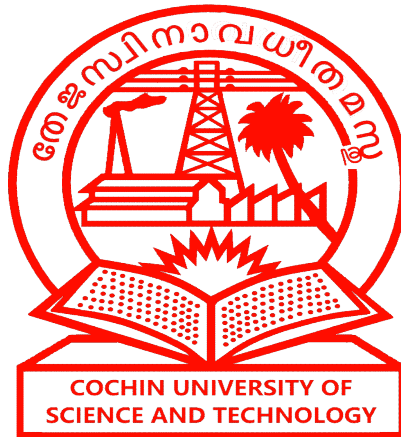
SCHOOL OF ENGINEERING

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

KOCHI-682022

APRIL 2023

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CERTIFICATE

*Certified that the Project report entitled “**SINE WAVE INVERTER WITH INSTANTANEOUS DIGITAL CONTROL**” is a bonafide work of **JAUHAR K** towards the partial fulfillment for the award of the degree of B.Tech in Electronics and Communication of Cochin University of Science and Technology, Kochi-682022.*

Project Guide
Dr Abdulla P

Head of Division
Dr Anju Pradeep

ACKNOWLEDGEMENT

First, we would like to thank ourselves for our effort and persistence towards our goal. We would like to express our thanks to a whole host of people who supported us in this endeavor of ours. Also we would like to thank each and every member of our respective families for their love and support.

Most importantly, we would like to thank Dr.Anju Pradeep, Head of the Division of Electronics and Communication Engineering and our project guide, Dr. Abdulla P for guiding and providing us with adequate facilities to help our task. Also our sincere gratitude to all the teachers who wholeheartedly supported us along with the Lab staff. We would like to thank all of our friends who were with us from the beginning till the end.

ABSTRACT

A sine wave inverter with instantaneous digital control, which is a type of inverter that uses advanced digital signal processing techniques to produce a high-quality sine wave output with extremely low harmonic distortion. The inverter converts DC input voltage to a high-frequency AC waveform using a switching circuit and processes the output waveform using a digital signal processor (DSP). The DSP uses advanced algorithms to generate a high-quality sine wave output with high efficiency, low harmonic distortion, and fast response time. This technology is ideal for applications that require high-quality power with minimal harmonic distortion, such as powering sensitive electronic equipment or medical devices.

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CHAPTER 1

INTRODUCTION

A sine wave inverter with instantaneous digital control is a type of inverter that uses advanced digital signal processing techniques to produce a high-quality sine wave output with low harmonic distortion. AC loads require constant or adjustable voltages at their input terminals. When such loads are fed by inverters, it's essential that output voltage of the inverters is so controlled as to fulfill the requirements of AC loads. This involves coping with the variation of DC input voltage, for voltage regulation of inverters and for the constant volts/frequency control requirement. There are various techniques to vary the inverter gain. The most efficient method of controlling the gain (and output voltage) is to incorporate pulse-width modulation (PWM) control within the inverters. The carrier based PWM schemes used for multilevel inverters is one of the most straight forward methods of describing voltage source modulation realized by the intersection of a modulating signal (Duty Cycle) with triangular carrier waveforms.

1.1 Scope

Inverters are used For many applications, as in situations where low voltage DC sources such as Batteries,solar panels or fuel cells must be converted so that devices can run Off of AC power. One example of such a situation would be converting Electrical power from a car battery to run a laptop, TV or cell phone.

1. Renewable energy: Sine wave inverters with instantaneous digital control can be used to convert DC power from solar panels or wind turbines into high-quality AC power that can be used by homes and businesses.
2. Backup power: Sine wave inverters with instantaneous digital control can be used as backup power sources for homes and businesses during power outages, providing a reliable and stable power supply.
3. Medical equipment: Sine wave inverters with instantaneous digital control can be used to power sensitive medical equipment that requires high-quality, stable power, ensuring that these devices operate reliably and safely.

4. Industrial applications: Sine wave inverters with instantaneous digital control can be used in a wide range of industrial applications, such as powering motors, pumps, and other equipment that requires a stable and reliable power source.
5. Telecommunications: Sine wave inverters with instantaneous digital control can be used in telecommunications equipment, such as cell towers, to provide a stable and reliable power supply that is critical for the operation of these systems.

Overall, the scope of a sine wave inverter with instantaneous digital control is broad and has the potential to be utilized in a wide range of applications where high-quality, stable power is required.

1.2 Project Outline

- a) We plan to produce a pure sinwave inverter with instantaneous control that can support upto 700VA of power, using a TI TMS320F280049C DSP processor.
Firstly, the required sinewave is produced using PWM module, with PWM frequency set at 10KHz. Varying the duty cycle with a sinusoidal reference produce sinusoidal PWM, or SPWM signal. The modulation scheme used here is Unipolar SPWM.
- b) The produced SPWM signal is fed to a H-Bridge to produce high power SPWM, using MOSFET switches.
- c) The output from H-Bridge is fed to transformer, which steps up the voltage from DC bus level to 230V level.
- d) Output from transformer is passed through an LC filter (Transformer inductance is used as L itself, no other inductance used), to filter out carrier wave and produce 50Hz AC output.
- e) Feedback from output is taken and processed in the DSP to stabilize the output using digital compensation techniques.

1.3 Limitations.

Sine wave inverters with instantaneous digital control have some limitations, including:

1. Complexity: The control circuitry required for instantaneous digital control is complex and requires advanced digital signal processing algorithms. This can increase the cost and complexity of the system.

2. Sampling rate: The performance of the inverter is dependent on the sampling rate of the control system. Higher sampling rates are required for improved accuracy and faster response times, which can be challenging to implement in high-power systems.
3. Harmonic distortion: Despite the instantaneous digital control, the output of the inverter may still have some harmonic distortion. This can be due to imperfections in the control system, non-ideal behavior of the power switches, or other factors.

1.4 System Specifications

230V inverter specs	
DC Input	12V
AC output	230V, 50Hz sinusoidal
Rating	700VA
Output filtering	LC
AC input	230v
PWM frequency	10Khz
Controller ISR frequency	100MHz
Controller used	TI TMS320F280049C
Compensation type	PR Compensator

CHAPTER 2

GETTING STARTED

2.1 Project Flow

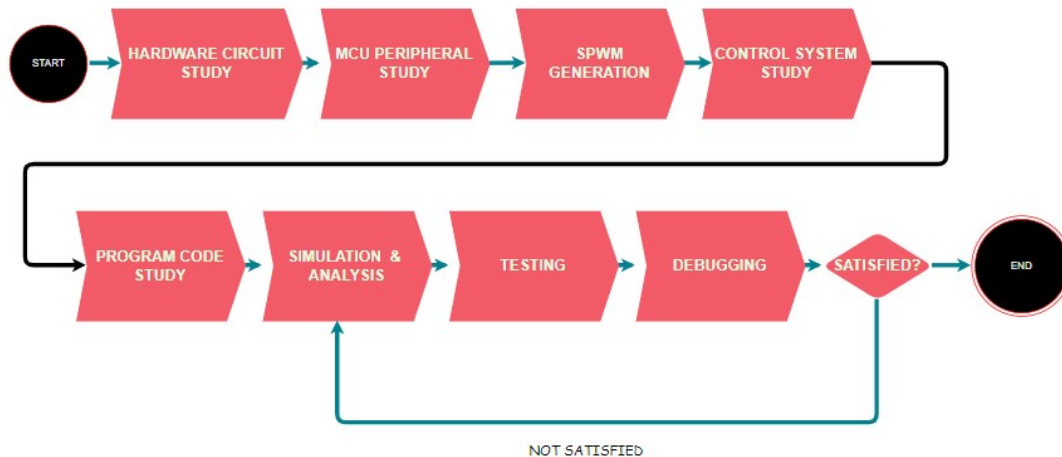


Figure 2.1 project flow planning

We divided the project flow to 8 stages below. Each level is important for the proceeding to next level. We purchased the hardware board containing the sampling circuits and H bridge. The compensation techniques and algorithms has to be implemented using DSP.

- 1) In Hardware circuit study, we began to learn the concepts and circuits of basic inverter hardware implementation.
- 2) After that, we studied about dsp used, which is TMS320F280049C DSP from Texas Instruments. Peripherals like ADC, Timers, ePWM and Interrupts were studied and related APIs were studied using Controller reference manuals and Datasheets.
- 3) SPWM generation schemes were studied.
- 4) Compensator schemes and designs were studied. Compensator types like average control, PI control and PR controls were studied.
- 5) TI provides sample designs and program codes along with other development and utility tools required. Those are accessible for everyone for free. The program code and algorithm were studied.

From following steps from 1-5, we can design our raw inverter, without any feedback control. The following steps are used for tuning and making it work.

- 6) Simulation and analysis of inverter. SFRA tool from TI was used for extracting the plant info from the actual hardware. The data from SFRA was used for making the simulation in Matlab and simulink.

Sine wave inverter with instantaneous digital control

- 7) Steady state and transition states of inverter was tested using DSO. The output has to respond to step load, step input, slow input variation and slow load variation.
- 8) In case of errors, or imperfect output, debugging the code using TI provided debugging tools, like memory browser, or hardware debugging using DSO.

If error is persisting, tune the system again and follow steps from 6-8.

2.2 System Block diagram

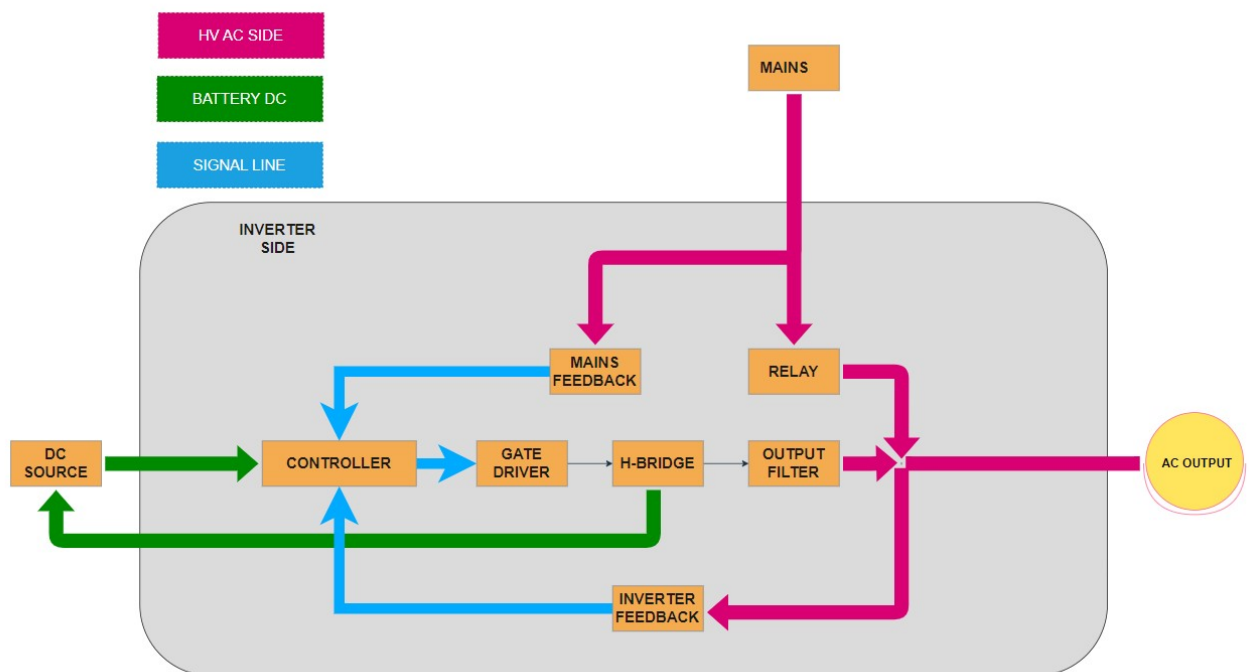


Figure 2.2 Inverter system block diagram

A block diagram for a sine wave inverter with instantaneous digital control typically includes the following components:

1. DC power supply: The DC power supply provides the input power to the inverter circuit. The voltage level of the DC supply determines the maximum output voltage of the inverter.

2. H Bridge circuit: The H bridge is responsible for converting the DC input power into an AC output waveform. It typically consists of power switches (e.g. MOSFETs or IGBTs) arranged in a configuration such as a full-bridge or half-bridge topology.'
3. Driver circuit: The driver circuit controls the switching of the power switches in the inverter circuit. It receives signals from the control circuit and uses them to turn the switches on and off at the appropriate times.
4. Control circuit: The control circuit is responsible for generating the signals that control the switching of the power switches. It typically includes a microcontroller or DSP that implements digital signal processing algorithms to generate the desired AC output waveform.
5. Output filter: The output filter is used to smooth out the AC output waveform and reduce harmonic distortion. It typically includes passive components such as inductors and capacitors.

2.3 Hardware circuit study

In getting started phase, its important to get familiar with the hardware and tools in hand.

Inverter hardware includes-

- 1) H bridge
- 2) OP Amps for sampling
- 3) Filtering Capacitor
- 4) Relays for Mains-Inv switching
- 5) Relays for Charging ON/OFF
- 6) Microcontroller control input ports
- 7) 5V regulator for controllers
- 8) High Voltage protection

2.3.1 Half-Bridge circuit

Half-Bridge VSI Figure 13 shows the topology of a Half-Bridge VSI, where two large capacitors are required to provide a neutral point N, such that each capacitor maintains a constant voltage $v/2$. Because the current harmonics injected by the operation of the inverter are low-order harmonics, a set of large capacitors (C^+ and C^-) is required. The duty cycle of

the switches is used to modulate the output voltage. The signals driving the switches must ensure some dead time to prevent shorting of the DC bus

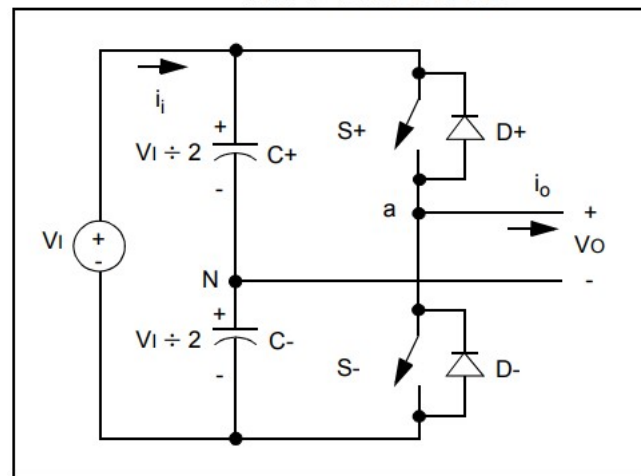


Figure 2.3 Half bridge circuit

2.3.2 Full-Bridge/H-bridge circuit

Figure 14 shows the topology of a Full-Bridge VSI. This inverter is similar to the half-bridge inverter; however, a second leg provides the neutral point to the load. Both switches S1+ and S1- (or S2+ and S2-) cannot be on simultaneously because a short circuit across the DC link voltage source V_i would be produced. To avoid the short circuit across the DC bus and the undefined AC output voltage condition, the modulating technique should ensure that either the top or the bottom switch of each leg is ON at any instant. The AC output voltage can take values up to the DC link value V_i , which is twice the value obtained with half-bridge VSI topologies. Several modulating techniques have been developed that are applicable to full-bridge VSIs. Among them, the best known are bipolar and unipolar PWM techniques.

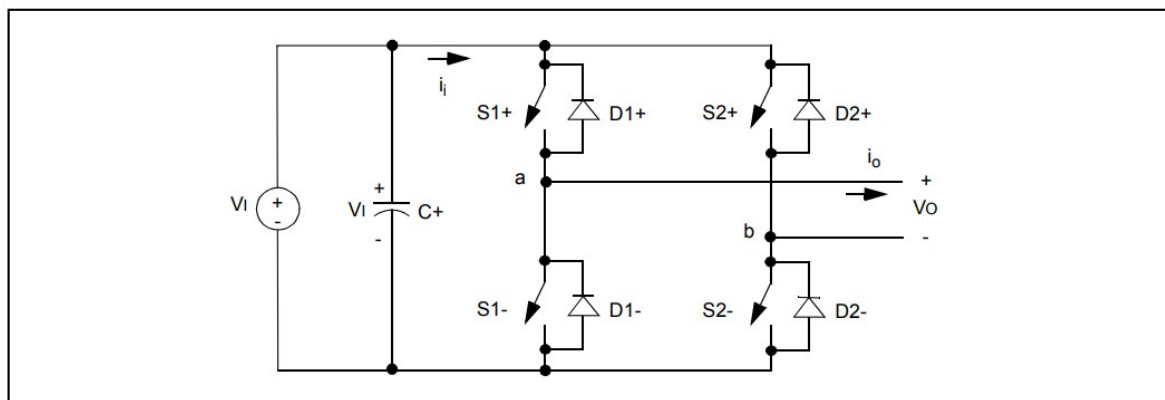


Figure 2.4 Full bridge circuit

2.3.3 Voltage Sensing

The AC output voltage is sensed differentially using resistor dividers and op amps, as shown in Figure 4. An offset voltage is added to the signal to enable measurement using the ADC, which can only convert positive voltages.

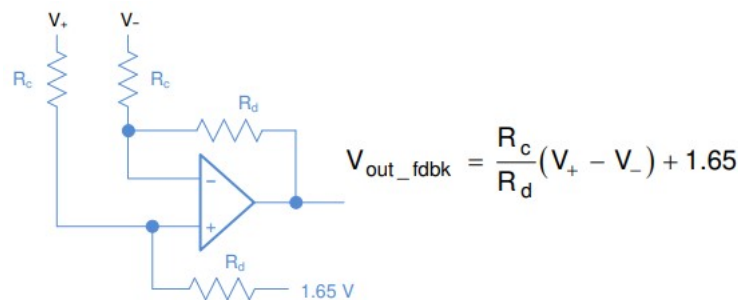


Figure 2.5 Voltage sense

2.4 Micro-controller study basics

We used TI TMS320F280049C 32bit DSP controller for our design. TI provides an ecosystem of development tools and libraries with which we can design and develop system more easily. Features of the mentioned chip includes,

- 100Mhz Clock frequency
- IEEE 754 single-precision Floating-Point Unit
- 3.3V
- Enhanced PWM capability
- 12 bit ADC, with 3.45 MSPS aquisition rate, with Integrated PPBs

TI provides example projects, with code and implementation support in many domains and we studied them for our project

2.4.1 ADC module

We studied features of ADC peripherals, and made reliable and accurate functions for reading ADC signals. ADC contains PPB block which corrects ADC offsets instantly. We Implemented algorithm for reading ADC result for multiple channels. We ADC triggered automatically with PWM signal, and subsequent channels are triggered in Round robin fashion. Results are generated and then interrupt occurs, and ADC result are processed to generate values for the next PWM cycle.

2.4.2 ePWM

TI provides ePWM (Enhanced Pulse Width Modulation) peripheral for generating PWM signals. ePWM contains

- Time-Base Submodule
- Counter Compare Submodule
- Action Qualifier Submodule
- Dead-Band Generator Submodule
- PWM Chopper (PC) Submodule
- Trip Zone Submodule
- Event Trigger Submodule
- Digital Compare Submodule

CHAPTER 3

OPEN LOOP IMPLEMENTATION

Open loop implementation simply means generation is the basic step for starting inverter design. There is no feedback. There is no compensation. Inverter produces pure sinewave output at no load condition. But significant variations occurs during loading. We generated Sine wave using unipolar SPWM technique, which adjusts the duty cycle of carrier wave according to a reference 50Hz sinewave.

Steps for generating sine wave in open loop is,

- 1) Create an algorithm for turning ON and OFF the required MOSFET's using PWM.
- 2) Adjust the duty cycles PWM carrier wave to follow a sinusoidal 50Hz reference signal. This PWM signals are phase shifted according to unipolar SPWM techniques and fed to corresponding switches in the H-Bridge.
- 3) Filtering the generated Unipolar SPWM output will get pure Sine wave

The algorithm was implemented using TMS320F280049C DSP from Texas Instruments, using their CCS IDE.

3.1 Unipolar SPWM Generation

In this scheme, the triangular carrier waveform is compared with two reference signals which are positive and negative signal. The basic idea to produce SPWM with Unipolar voltage switching is shown in Figure 2.9. The different between the Bipolar SPWM generators is that the generator uses another comparator to compare between the inverse reference waveform V_- . The process of comparing these two signals to produce the Unipolar voltage switching signal is graphically illustrated in Figure 2.10. In Unipolar voltage switching the output voltage switches between 0 and dc V , or between 0 and dc V_- . This is in contrast to the Bipolar switching strategy in which the output swings between dc V and dc V_- . As a result, the change in output voltage at each switching event is halved in the Unipolar case from $2V_{dc}$ to V_{dc} . The effective switching frequency is seen by the load is doubled and the

Sine wave inverter with instantaneous digital control

voltage pulse amplitude is halved. Due to this, the harmonic content of the output voltage waveform is reduced compared to Bipolar switching. In Unipolar voltage switching scheme also, the amplitude of the significant harmonics and its sidebands is much lower for all modulation indexes thus making filtering easier, and with its size being significantly smaller.

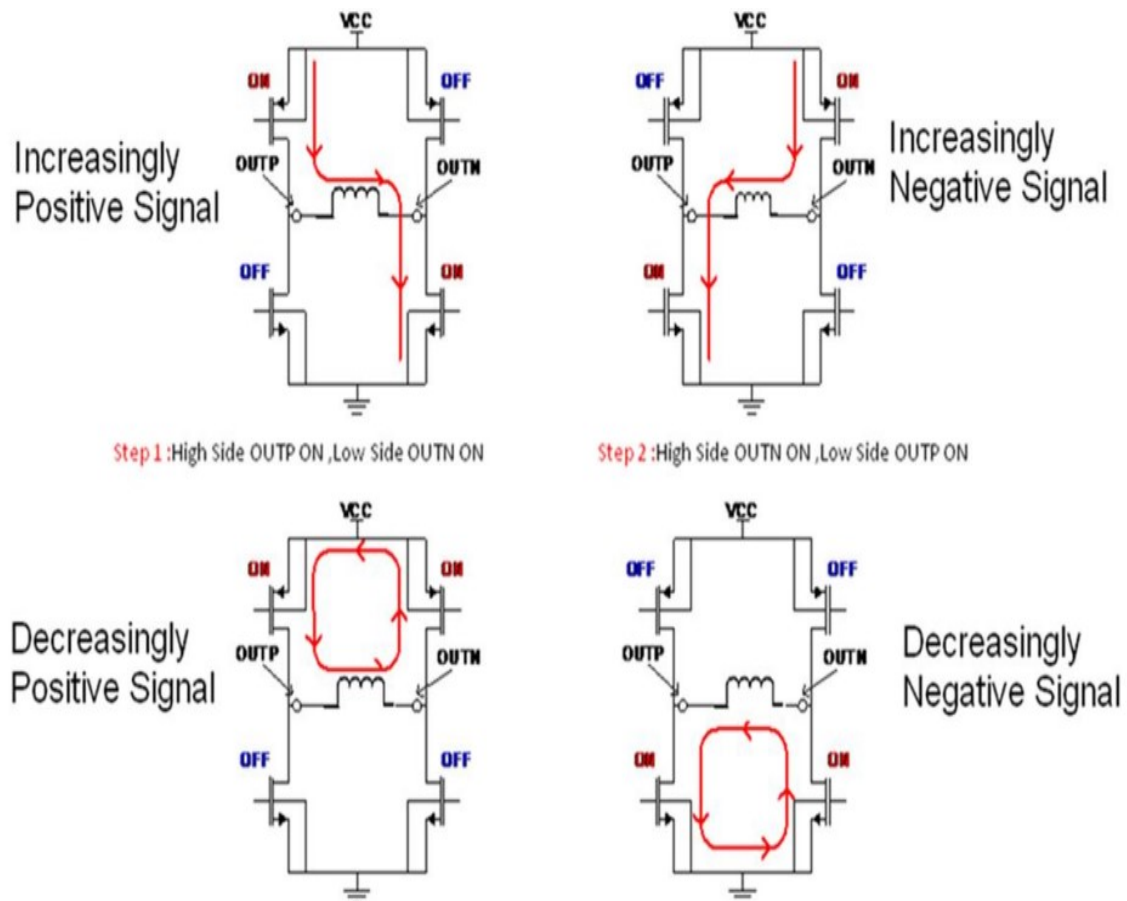


Figure 3.1 Unipolar spwm switching

For Unipolar SPWM,

- 1) corresponding Higher and Lower switches (FETs) should be complementary. ie, 1_H and 1_L should be complementary
- 2) Switches parallel to each other should be 180 degree out of phase. ie, 1_H and 2_H should be 180 degree phase shifted.
- 3) 2 PWM channels used for controlling the high side switches. Their complement is used for low side switches.
- 4) Both channels should be synchronised.

Short circuit occurs when both switches in same side are turned ON simultaneously.

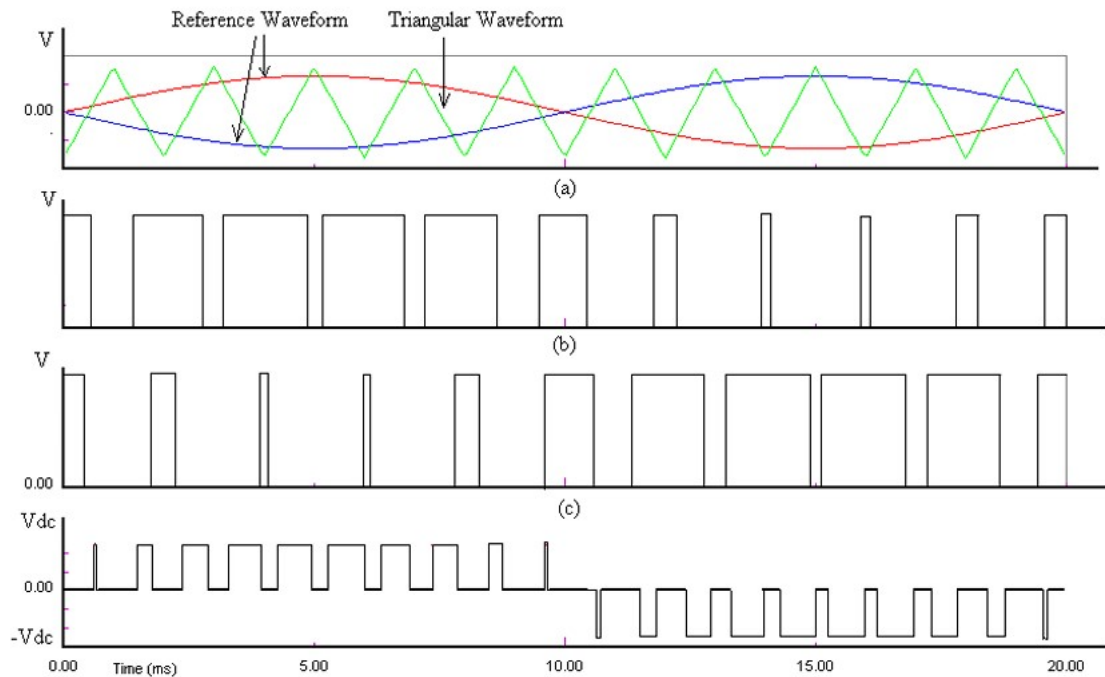


Figure 3.2 Unipolar SPWM signals. a) signal on High side transistor 1, b) signal on High side transistor 2. The signals for low sides are taken as complement of high side transistors.

3.2 RED and FED

TI ePWM module provides Deadband submodules which can provide Falling edge and Rising edge delays which provides a gap between the turning OFF of one switch and the turning ON of its complementary switch. RED stands for Rising Edge Delay and FED stands for Falling Edge Delay. Without FED and RED, during transitions, conduction states of two switches in same side overlaps. This causes a short circuit path, and high current flows, which then results in power losses and could end in damaging the circuit. 10 μ s delay is used in our project.

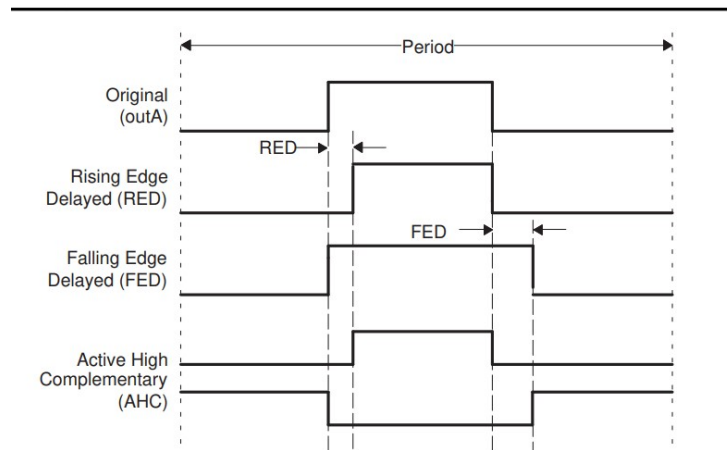


Figure 3.3 FED and RED

3.3 Implementation

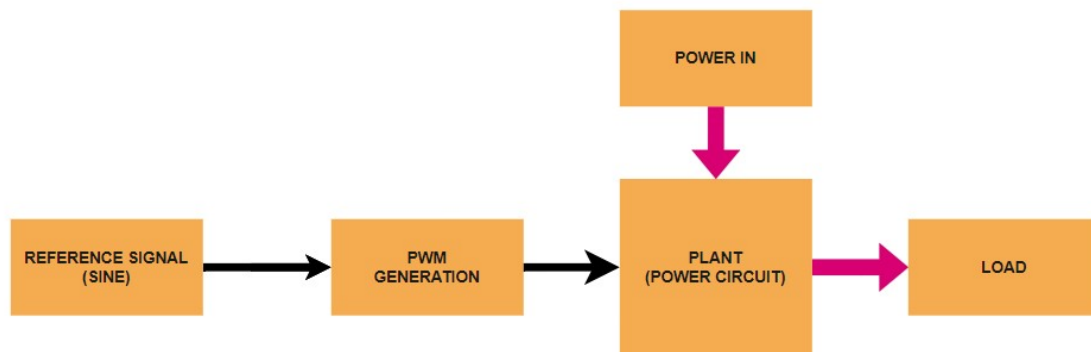


Figure 3.4 Open loop system block diagram

- 1) PWM carrier wave of 10Khz signal is setup
- 2) Timer interrupt set at every 10Khz
- 3) Sine value for that time is generated using each interrupt.
- 4) Generated sine value is used to vary the pwm duty cycle accordingly to implement unipolar spwm.
- 5) This is fed to Plant, ie, the inverter hardware, which then outputs the sine wave at 50hz, 230V.

3.4 Open Loop output

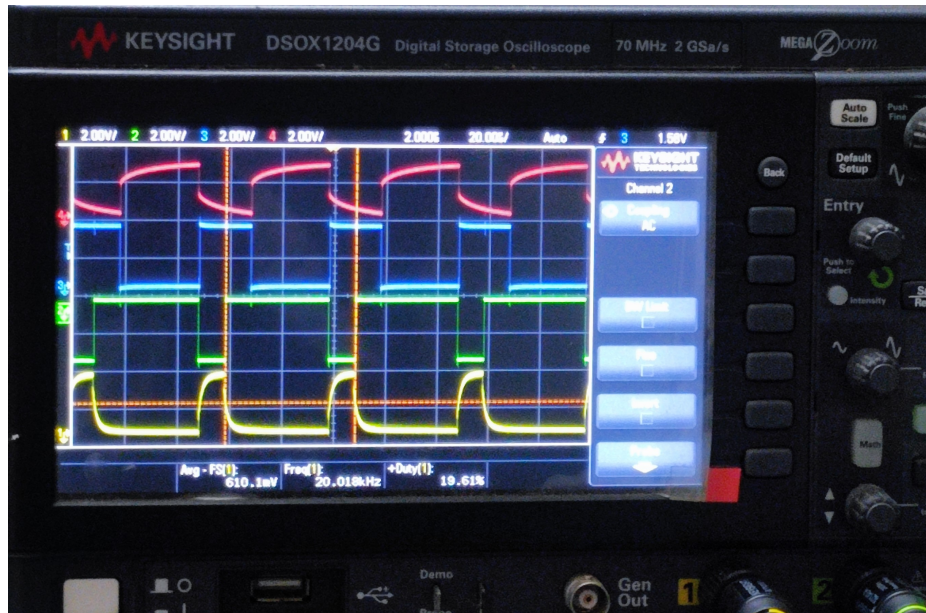


Figure 3.5 MOSFET input signals

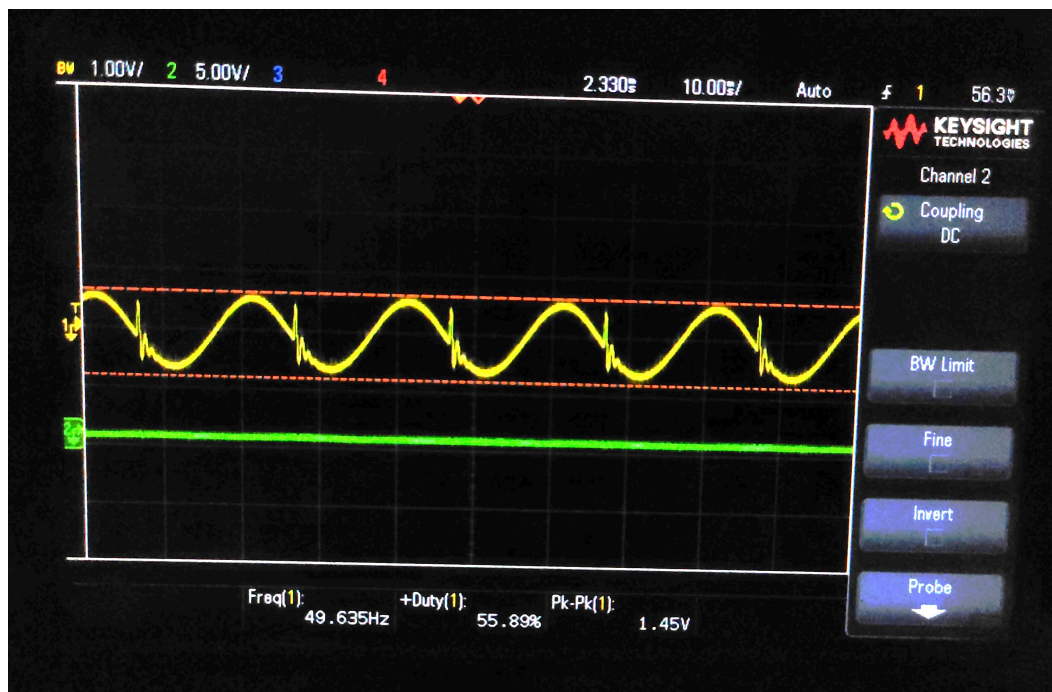


Figure 3.6 Inverter output

230V, 50Hz sinewave is obtained at the output.

CHAPTER 4

CLOSED LOOP IMPLEMENTATION

Adding a feedback for reading the output response of the system and then compensating it using a controller makes the system more stable and can respond to Load and Line variations. We learnt and tried different feedback methods including Average Control, Instantaneous PI control and Instantaneous PR Control. Of the 3, we got the best response from PR compensator, due to its fast response to step response as well as suppressing harmonics.

4.1 Controller selection

4.1.1 Average Mode Control

Here the average, or DC value of an individual Half cycle is taken using successive sampling and addition of ADC values, and then taking their mean after every zero crossings. The upside is, Average control is easy and produce reliable output. But the downside is, it is cycle by cycle correction. It takes about 15-20 signal cycles for correcting step errors. It requires accurate Zero crossing detection.

4.1.2 Instantaneous PI Control

Probably the most common type of controller in industrial power electronics is the “PI” (Proportional - Integral) controller. It has Proportional and Integral controllers. Proportional Controller provides fast transient responses, while Integral controller removes steady state errors for a set value.

But PI controller produces a steady state error when following a sinusoidal reference, because whenever the PI controller compensates for the error, the reference has already changed. To address this problem, Instantaneous PR Compensator is used.

4.1.3 Instantaneous PR Control

PR controller stands for Proportional Resonant Controller. Ideal PR controller provides infinite gain at Resonant frequency and Zero gain at other frequencies so that there is no

steady state error. For practical controllers, there will be a fixed gain at resonant frequency. The bandwidth can be adjusted by changing the cutoff frequency.

$$G_{AC}(s) = \frac{Y(s)}{E(s)} = \frac{2K_i s}{s^2 + \omega^2}$$

$$G_{AC}(s) = \frac{Y(s)}{E(s)} = \frac{2K_i(\omega_c s + \omega_c^2)}{s^2 + 2\omega_c s + (\omega_c^2 + \omega^2)}$$

$$\approx \frac{2K_i \omega_c s}{s^2 + 2\omega_c s + \omega^2}$$

Where, ω_c is cutoff frequency and ω is the resonant frequency. K is the proportional gain. It provides high attenuation at unwanted frequencies and high gain at resonant frequencies. Also it provides good phase and gain margins.

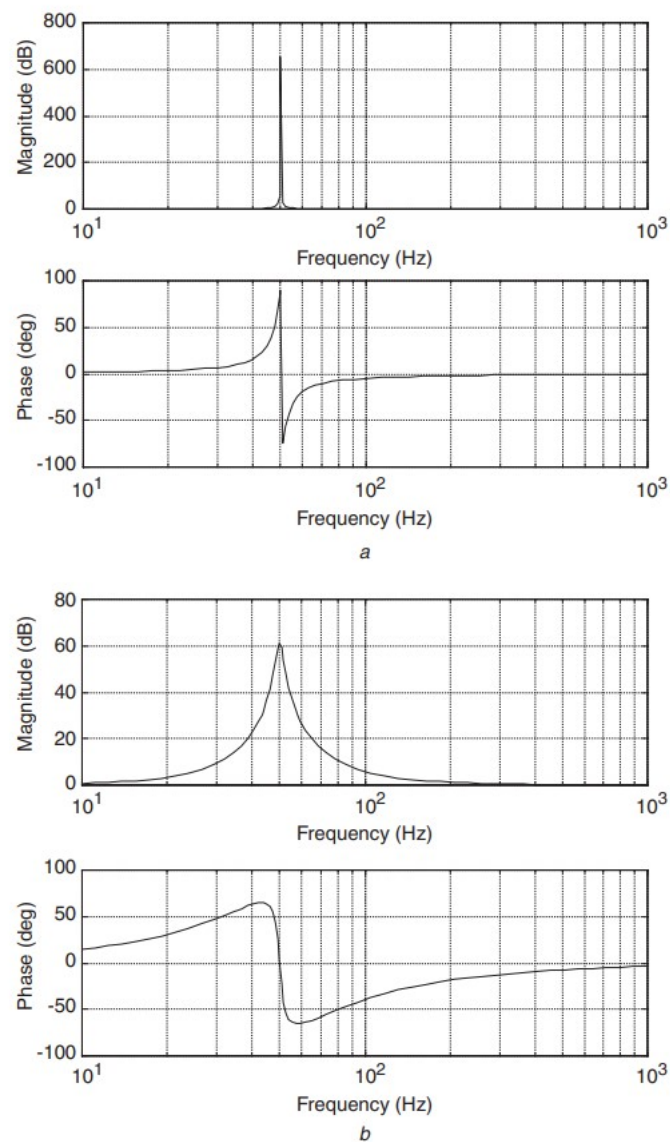


Figure 4.1 PR controller tuned at 50Hz. a) Ideal PR, b) practical PR

For our inverter, we considered fundamental frequency as 50Hz, and also considered its three harmonics -1st, 3rd and 5th harmonics. TI Provides APIs for implementing 2P2Z Controller in DCL library. TI example code uses 2P2Z Controller for implementing PR Controller. coefficients has to be calculated externally.

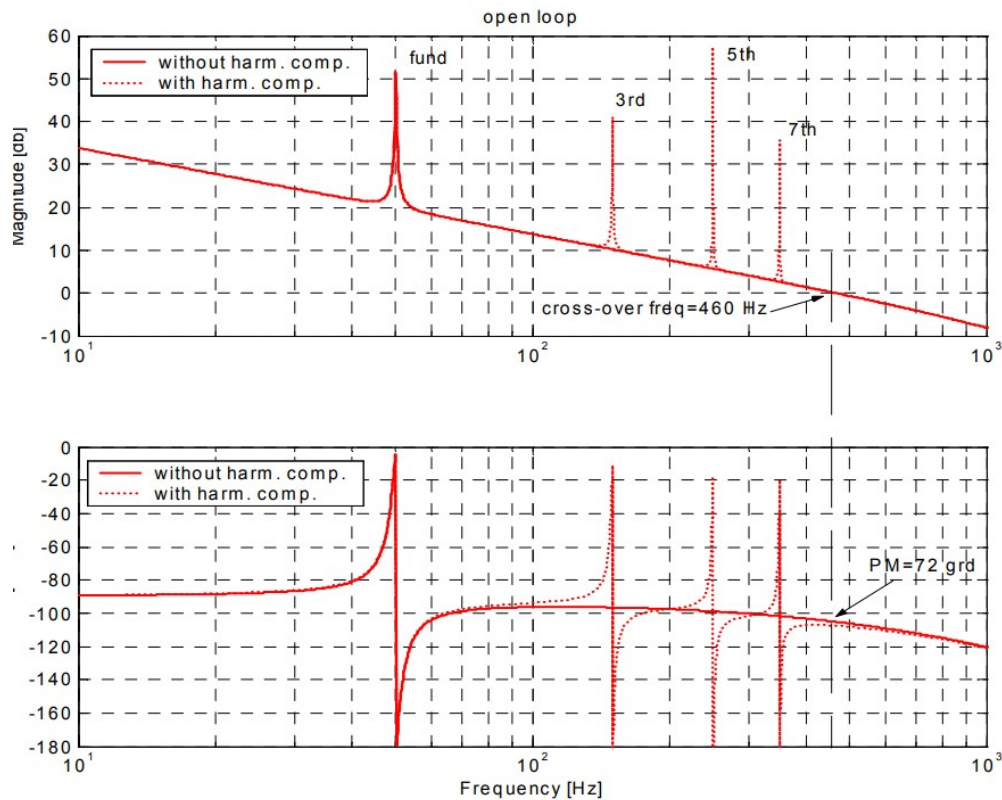


Figure 4.2 PR compensator with harmonics.

4.2 SFRA

Texas Instruments' software frequency response analyzer (SFRA) library is designed to enable frequency response analysis on digitally controlled power converters using software only and without the need for an external frequency response analyzer. The optimized library can be used in high frequency power conversion applications to identify the plant, the closed loop and the open loop gain characteristics of a closed loop power converter, which can be used to get stability information such as gain margin, phase margin and open loop gain crossover frequency, to evaluate the control loop performance.

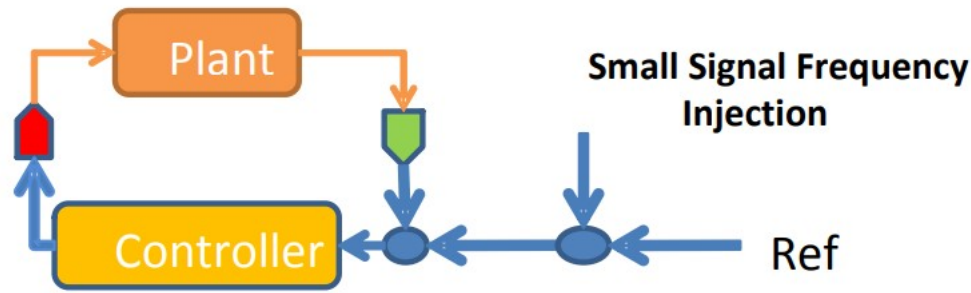


Figure 4.3 SFRA block diagram

The SFRA library is based on sinusoidal injection principle, where the assumption is that the injection amplitude causes very small deviation to the normal operating point of the converter. The SFRA library can be integrated into the control code of the power converter, this document details the steps to do so. All computations for the GH, H and CL calculations are done on the MCU and the entire arrays of the GH, H and CL magnitude and phase response are stored on the controller. For visualization purpose, a GUI is also provided with the tool. This GUI can be used to start a frequency response sweep, observe progress of the SFRA sweep and analyze the GH, H and CL plots generated after the FRA sweeps.

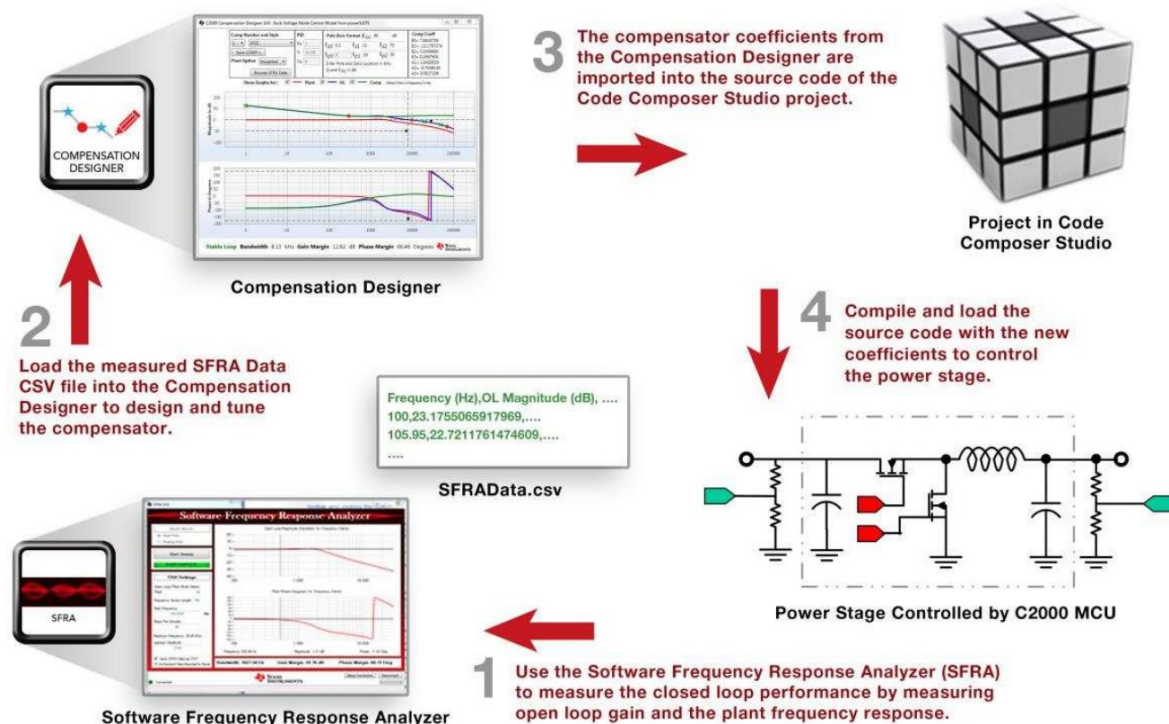


Figure 4.4 SFRA analysis

TI's software frequency response analyzer along with the Compensation Designer provides a methodology to tune power supplies in a systematic way and enables quick and easy frequency

response analysis for power converters without the need of external connections and equipment, as shown in Figure 2. Since no external connections are used, the SFRA can be run repeatedly to periodically assess the health of the power converter and get diagnostic information.

4.2.1 SFRA Measurement Output

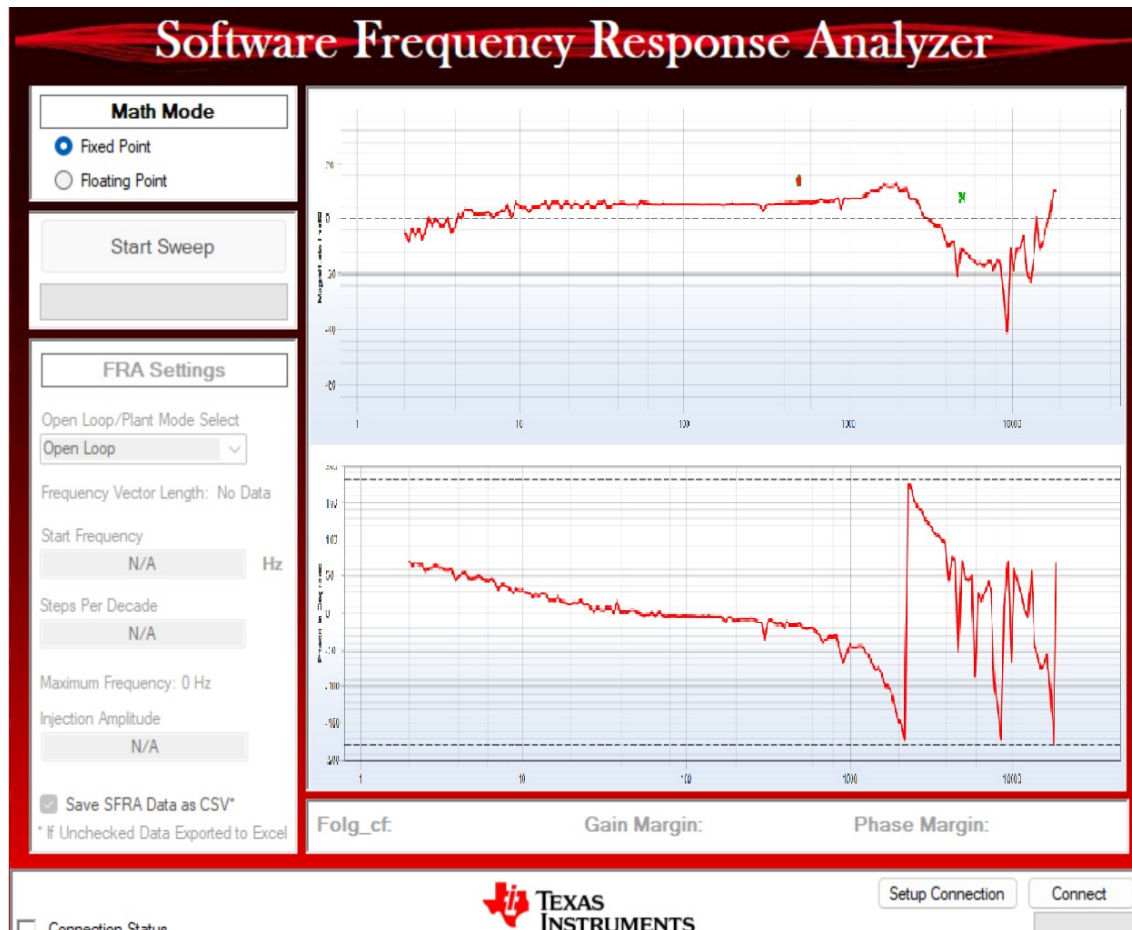


Figure 4.5 SFRA output

From the above figure, it shows the measured transfer characteristics of the plant. This data are used for simulation and tuning purpose.

4.3 Physical Modelling

Creating physical model of the system helps to understand the system gains and paths. It is important for scaling values in the program code and simulations. The total gain in a closed loop should be zero to make it stable.

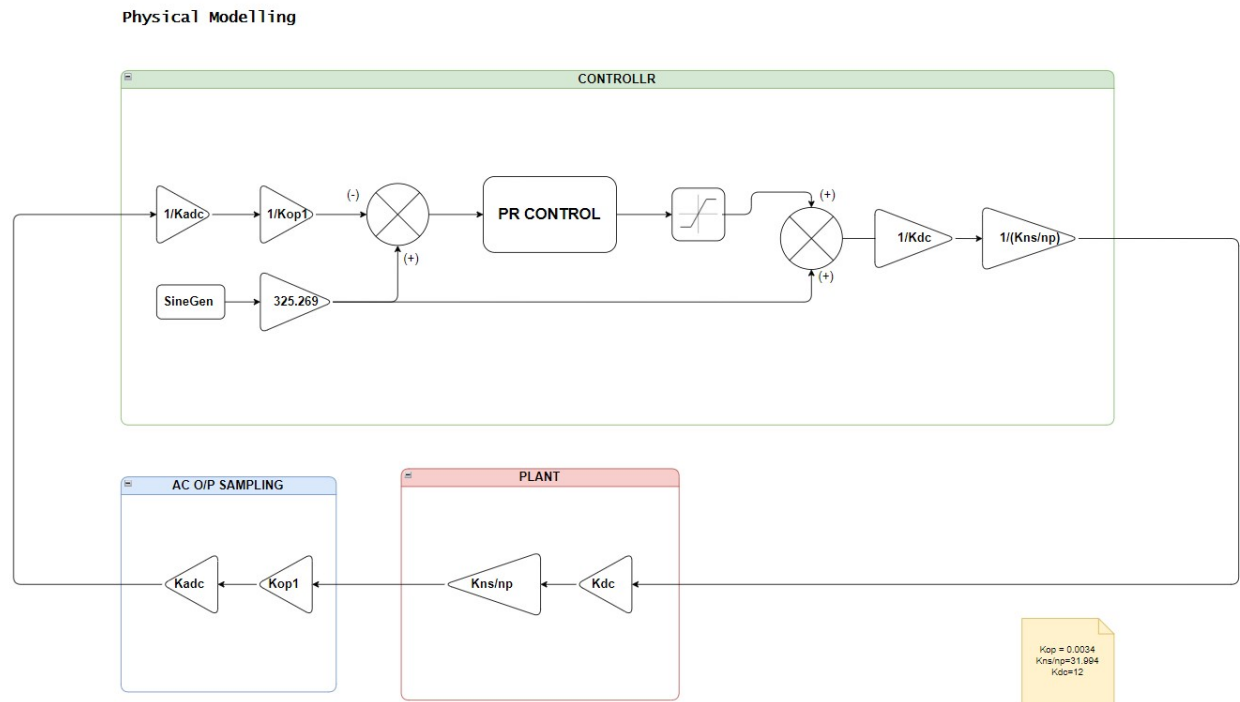


Figure 4.6 Physical modelling

4.4 Simulink Model

We used data from SFRA to create an approximate plant model for Simulink. Also TI provides Simulink blocks for their compensator models. We combined them together for making the closed loop system to simulate their responses. Coefficients for compensator model requires resonant frequency, bandwidth, Proportional and integral gains to calculate. These were given to a MATLAB script, which returned the coefficient values.

$$u(k) = b_0 e(k) + b_1 e(k-1) + b_2 e(k-2) - a_1 u(k-1) - a_2 u(k-2)$$

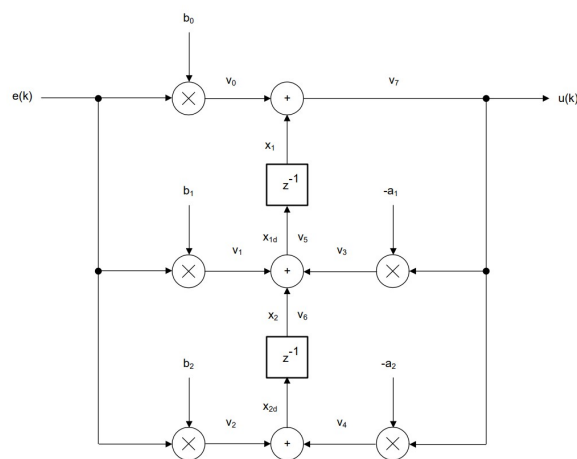


Figure 4.7 DF22 implementation

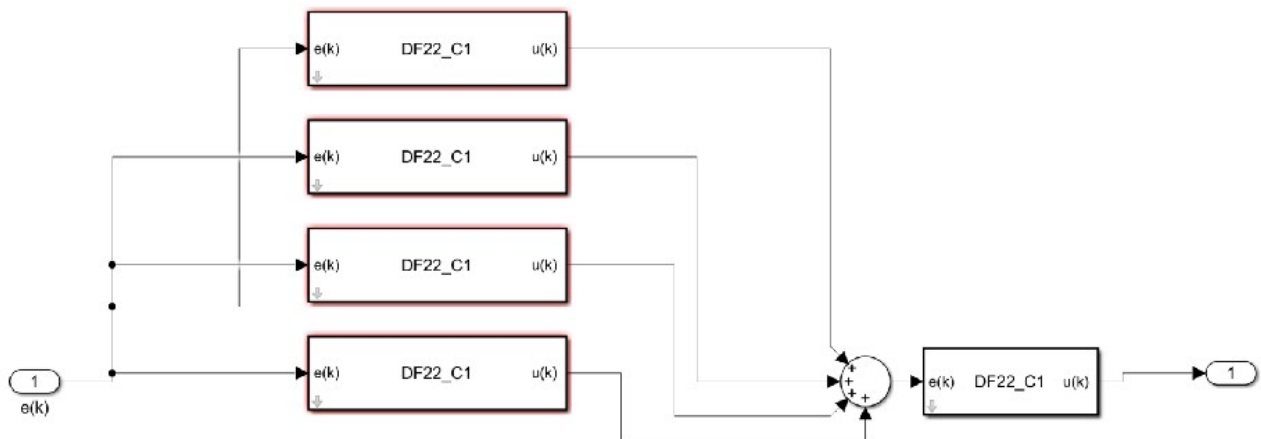


Figure 4.8 PR Controller implementation

The above figure shows the compensator implementation. It Contains 4 DF22 compensators tuned to fundamental frequency (50Hz), and odd Harmonics 150Hz, 250Hz, and 350Hz

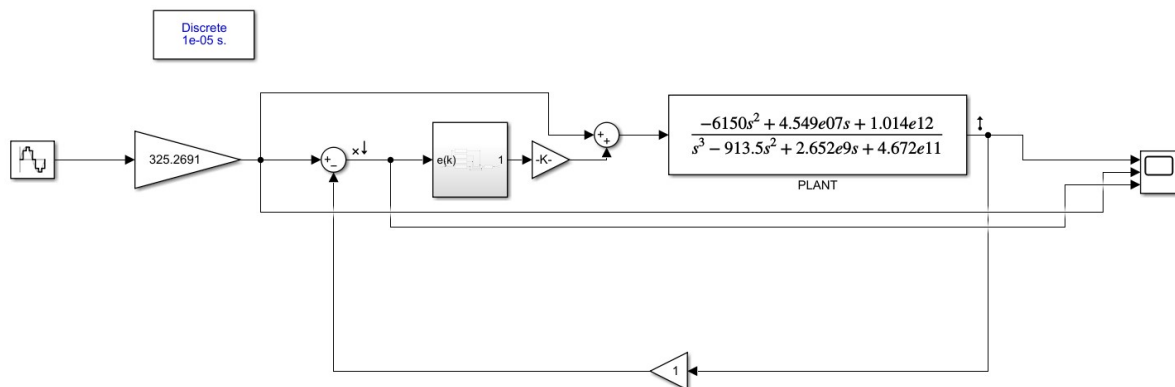


Figure 4.9 system simulink block diagram

The above figure shows the completed system block design.

4.4.1 Simulink Modelling - Simulation Output

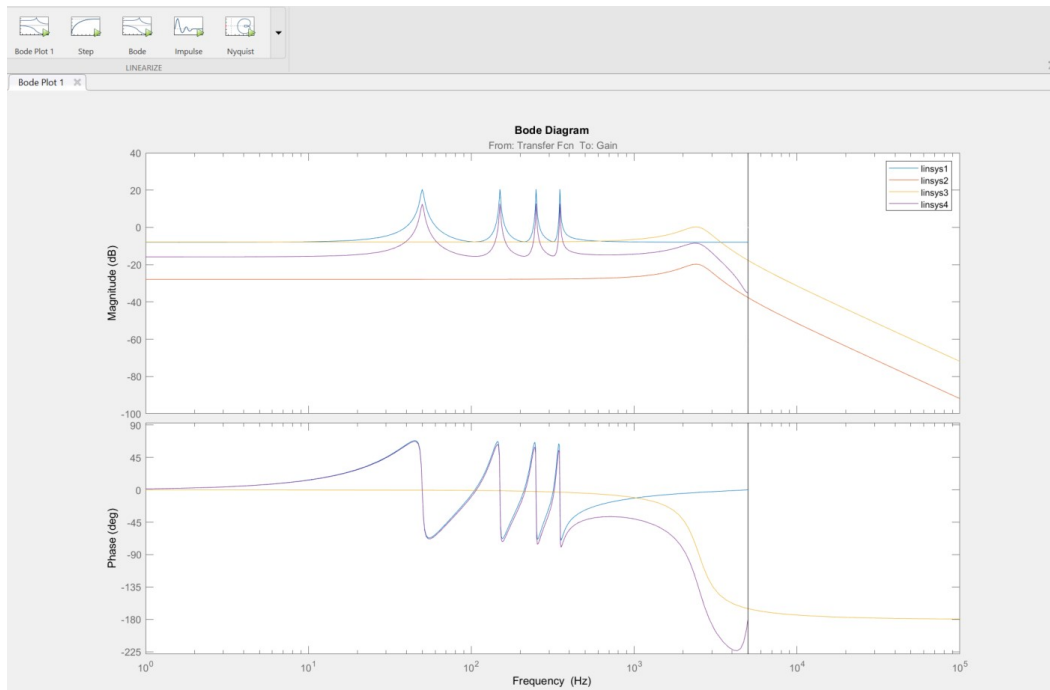


Figure 4.10 simulation output

The simulation output shows the frequency response for different coefficient values. The plot with highest gains at resonance and low gains at other frequencies, while also providing sufficient phase margins are selected for testing.

4.5 OUTPUT

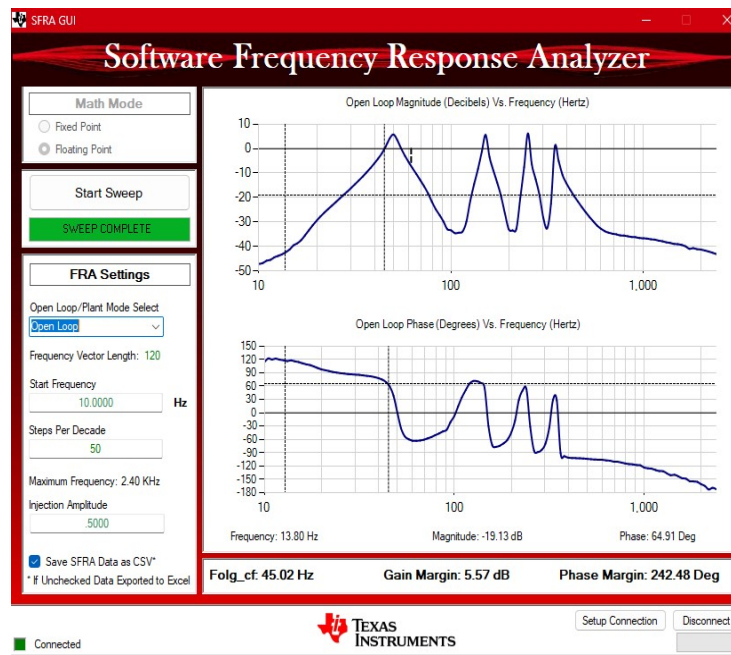


Figure 4.11 Open loop output

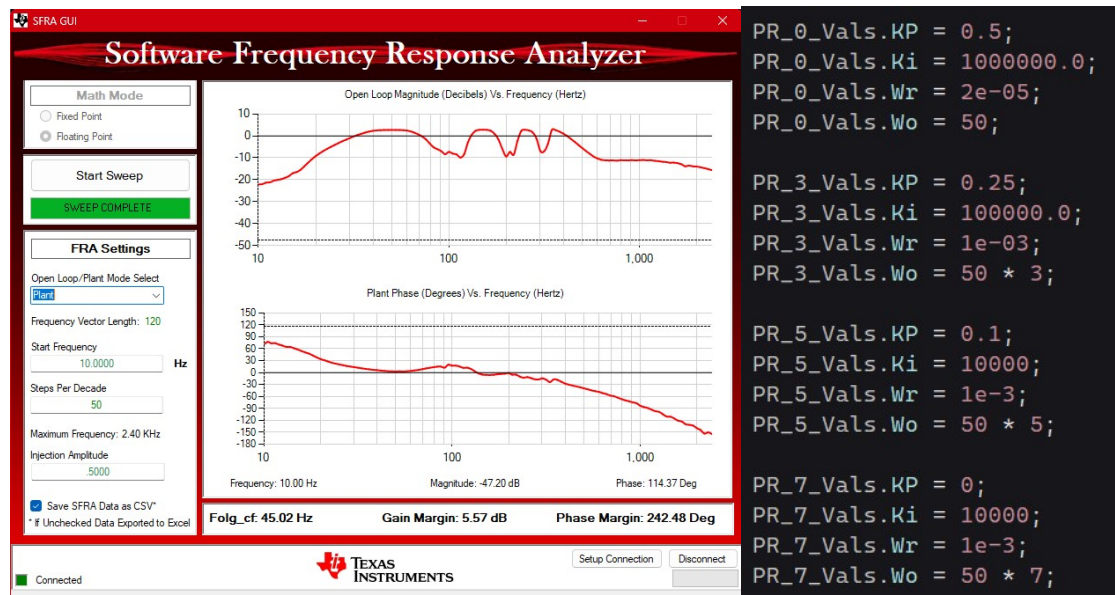


Figure 4.12 closed loop output, with the parameter values.

The above figures the open loop and closed loop transfer function of the system using SFRA, after coefficient values were added.

Sine wave inverter with instantaneous digital control

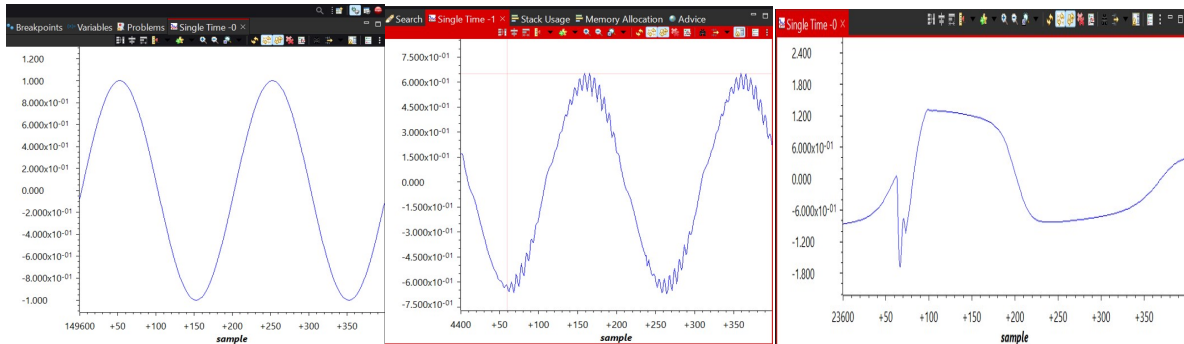


Figure 4.13 reference signal, No load output, Loading with 60W bulb

The above figures show the output from the system. It shows small oscillations at no load and severe issues when 60W bulb is connected. After a bit of troubleshooting, we found that the issue when load is connected is due to the current limit of DC supply used during testing. We swapped it for a battery instead. After another few rounds of testing and tuning, we got the result.



Figure 4.14 OUTPUT 1

In the above picture, the green probe is connected to the switch, and the yellow probe is connected to the output. So we can see that when the switch is pressed, there is a sudden decrease in the output amplitude, and then the compensator comes into action, which then corrects the output after a few cycles. Here there are small fluctuations at the output. So small tuning is required.

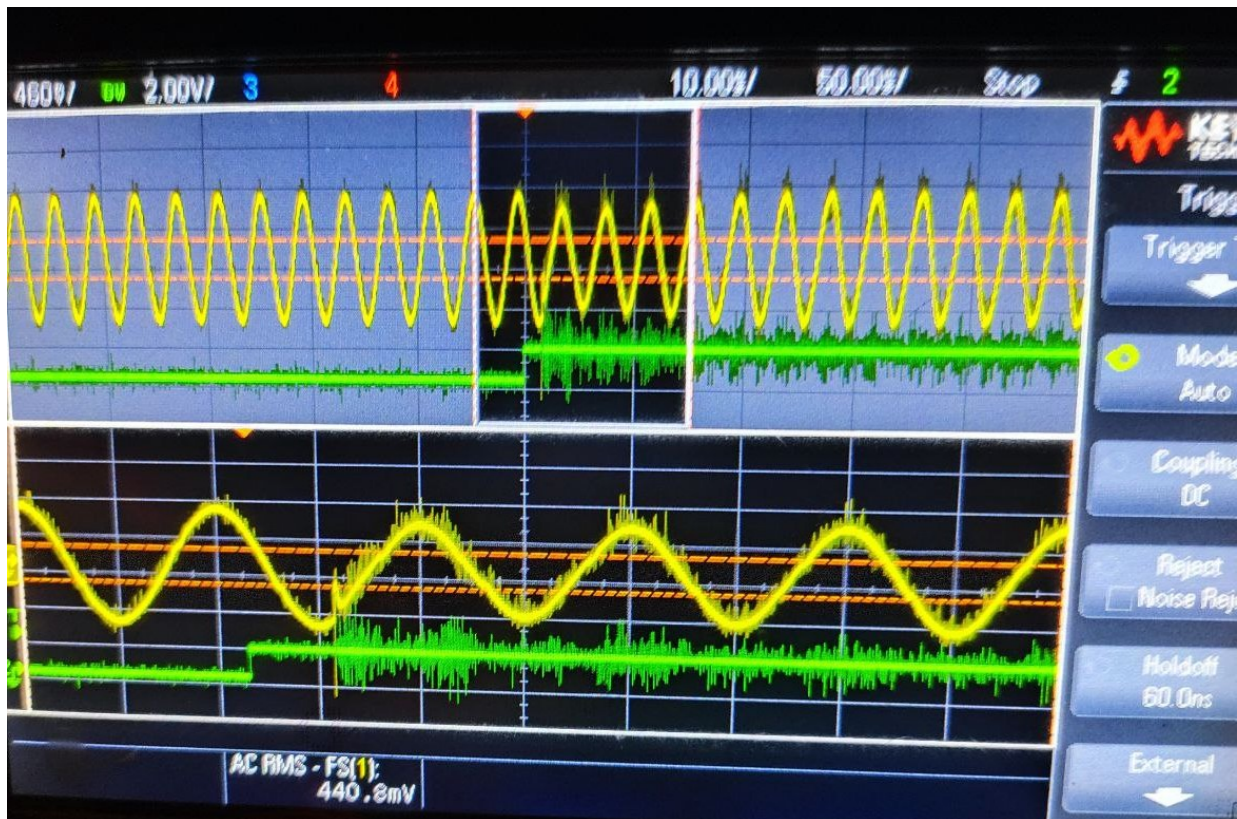


Figure 4.15 Final Output

This is the final result obtained after further tuning. There is still a decrease in output when step load is connected. But the compensator corrects it within 2-3 cycles. Also there is no unwanted oscillations. On summary, We produced the output we were looking for.

CHAPTER 5

STATE MANAGEMENT AND SYSTEM ALGORITHM

A high level block diagram of Inverter is shown in figure. The software is broadly classified into two parts

1. Inverter State Machine
2. User Interface software

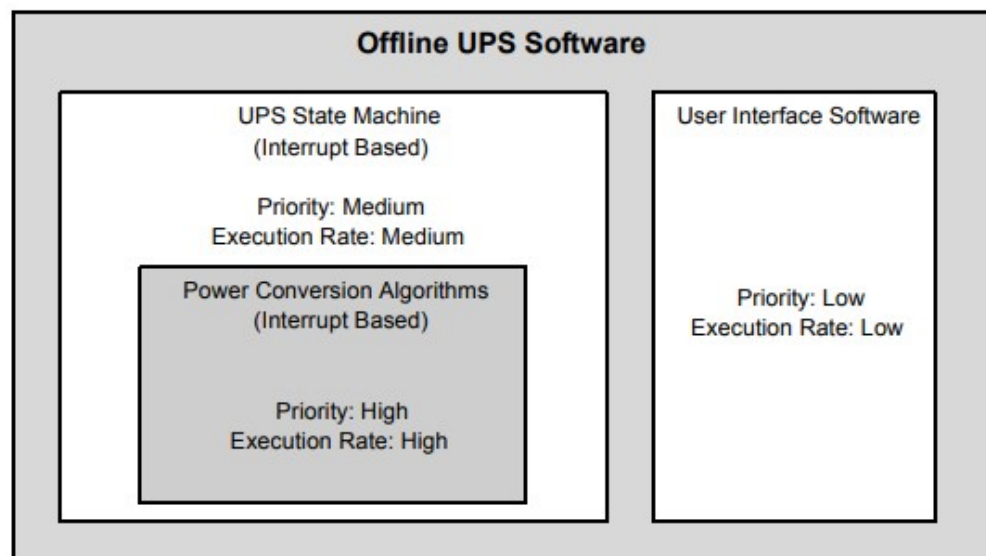


Figure 5.1 high level block diagram

5.1 Inverter State Machine

The Inverter software implements a state machine to determine the mode of operation for the system. The state machine is executed once every 100 μ s inside a timer Interrupt Service Routine (ISR). The state machine configures the on-chip peripherals to execute the correct power conversion algorithms. During normal operation of the inverter, the state machine configures the system peripherals to execute the correct power conversion algorithms as determined by the system state. When a power failure occurs, the Inverter state machine initiates a switchover sequence from Battery Charger mode to Inverter mode. When the AC mains is detected again, the state machine executes the switchover from Inverter mode to Battery Charger mode. These switchover functions must be executed in as little time as possible to ensure uninterrupted power to the load. The Battery Charger mode and Inverter

mode are the two normal operating modes of the system. There are two other modes of operation, namely System Startup and System Error.

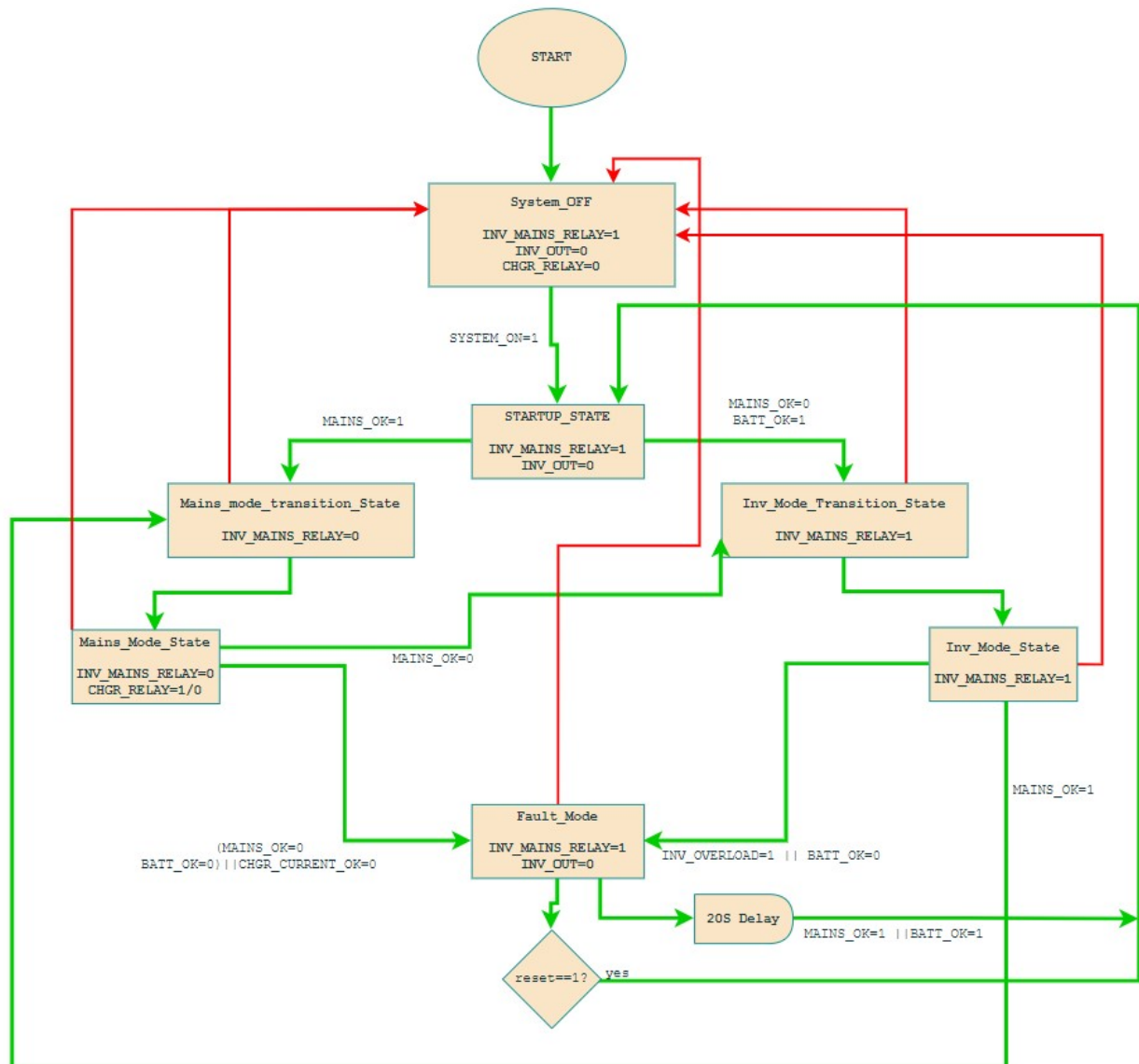


Figure 5.2 Inverter State machine

The above figure shows the different States used and the transitions between them.

5.2 User Interface

We used 16x2 LCD display and 3 push button switches for user interface. The switches were read using a single pin using ADC Multiplexing.

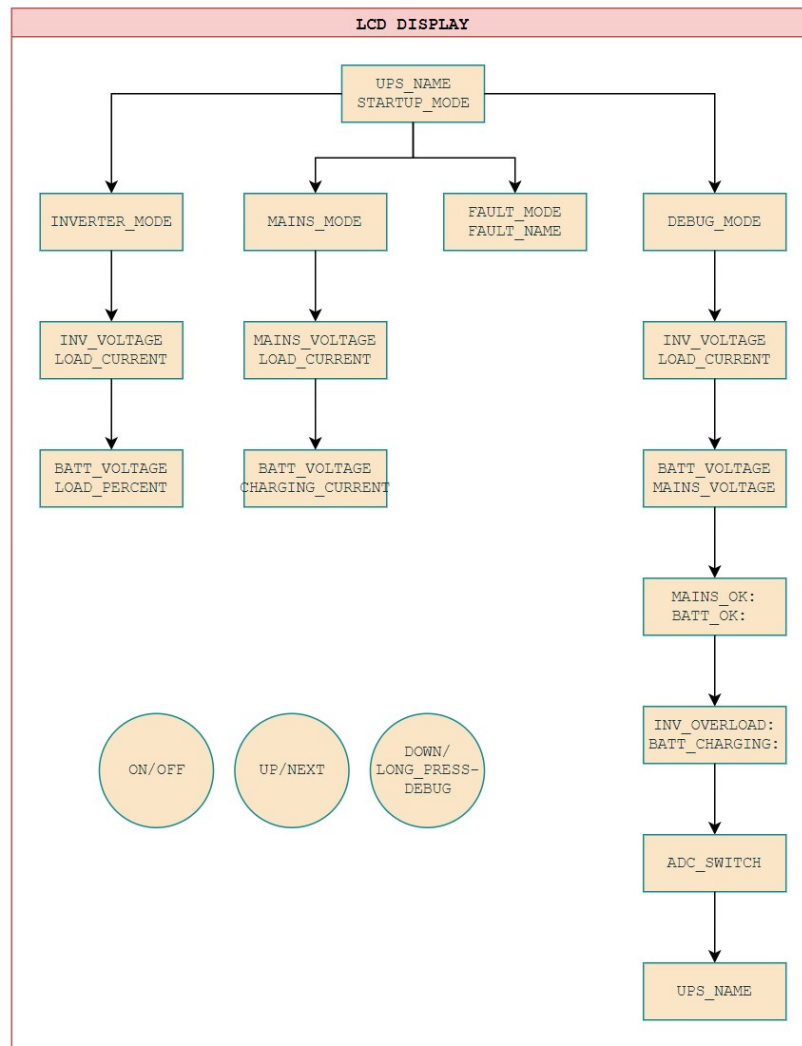


Figure 5.3 LCD UI

5.3 Adding final features- Soft Start, High Current Protection, DC bus voltage Protection

- Soft start helps to limit transient voltages, by providing a smooth ramping up.
- High current protection: Turning off inverter when the current limit exceeds a given value.
- DC bus voltage protection: Turning off inverter when DC bus voltage is Lower than 10V or higher than 14Volts.

CHAPTER 6

CONCLUSION

In conclusion, we have successfully designed, built and tested a pure sine wave inverter with an instantaneous PR controller that can provide a stable 230V output. By using TI's TMS320F280049C Digital Signal Processor, we were able to implement the necessary control algorithms and conduct rigorous testing with varying loads and lines. Our tests showed that the compensator was able to handle a step load of up to 60W within 2-3 cycles, or approximately 40-60mS. Through this project, we gained valuable knowledge about different types of inverters, compensation methods, and industry standard tools such as TI CCS, C2000Ware library, SFRA, Matlab and Simulink. It's worth noting that inverters are an important component used in a wide range of applications including renewable energy systems, electric cars, homes, and industries. Overall, this project has been a great learning experience for us and we are excited to apply our newfound knowledge to future projects.

CHAPTER 7

REFERENCES

1. TI inverter reference design

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4. C2000ware SFRA

https://www.ti.com/lit/ug/spruhz5a/spruhz5a.pdf?ts=1682478298243&ref_url=https%253A%252F%252Fwww.google.com%252F

5. Design and Analysis of Controller with Selective Harmonic compensator for Grid Connected PV System

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